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1. Mutagenesis and its types?

Mutagenesis is a process to change the genetic information of an organism It can occur naturally, or as a result of exposure to mutagens or induced experimentally in laboratory, Three different methods of site-directed mutagenesis have been devised i). Cassette mutagenesis ii). Primer extension iii). Procedures based on PCR.

Cassette mutagenesis A synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the wild-type gene.

Primer extension: the single primer method It involves priming in vitro DNA synthesis with a chemically synthesized oligonucleotide (7-20 bp) that carries a base mismatch with the complementary sequence.

PCR methods for site directed mutagenesis Single bases mismatched between the amplification primer and the template become incorporated into the template sequence as a result of amplification.

DNA sequencing: Benefits and Applications

DNA sequence information is a prerequisite for planning any substantial ~~manipulation~~ of the DNA

Information is useful to various fields i.e. molecular & evolutionary biology, metagenomics, medicine, forensics etc.

Techniques for DNA sequencing became available in the late 1970s

Maxam-Gilbert method:

This method for DNA sequencing makes use of chemical reagents to bring about base-specific cleavage of the DNA

3. RT-PCR and Multiplex PCR?

RT-PCR:

Reverse transcription followed by the PCR (RTPCR) leads to the amplification of RNA sequences in cDNA form

 RT-PCR is a sensitive means for detecting, quantifying and cloning specific cDNA molecules. Major screening strategies involve

1). Genetic methods

2). Sequence-dependent screening

3). Screening expression libraries

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Multiplex PCR:
Multiplex PCR is used to amplify several different DNA sequences simultaneously.
Multiplex PCR is a widespread molecular biology technique for amplification of multiple targets in a single PCR experiment. In a multiplexing assay, more than one target sequence can be amplified by using multiple primer pairs in a reaction mixture. As an extension to the practical use of PCR, this technique has the potential to produce considerable savings in time and effort within the laboratory without compromising on the utility of the experiment.

real-time PCR
The main advantage of **real-time PCR** over **PCR** is that **real-time PCR** allows you to determine the initial number of copies of template DNA (the amplification target sequence) with accuracy and high sensitivity over a wide dynamic range.

In conventional PCR, the amplified DNA product, or amplicon, is detected in an end-point analysis. In real-time PCR, the accumulation of amplification product is measured as the reaction progresses, in real time, with product quantification after each cycle. First, amplification reactions are set up with PCR reagents and unique or custom primers. Reactions are then run in real-time PCR instruments and the collected data is analyzed by proprietary instrument software.

5. Disadvantage of PCR?

Setting up running required high technical skills.
High equipment cost.
High Test Cost
High sterile environment should be provided. (DNA contamination)

6. Ortholog?

Orthologs are homologous genes in different organisms that encode proteins with the same function and which have evolved by direct vertical descent.

7. Four Step of Cloning Strategy?

Any cell based cloning procedure has four essential parts i). A method of generating DNA fragment for cloning ii). A reaction that inserts that fragment into the chosen cloning vectors iii). A means for introducing that recombinant vector into a host cell wherein it is replicated iv). A method of selecting recipient cells that have acquired the recombinant

8. Spotting DNA?

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Spotted DNA arrays are produced by printing DNA samples on treated microscope slides. It is made by transferring or spotting DNA clones or PCR product individually onto a solid support where they are immobilized.

9. Why taq DNA polymerase use in PCR?

Taq polymerase is a thermostable DNA polymerase isolated from thermophilic bacterium Thermus aquaticus and is used in PCR to amply DNA fragments.

10. How Fragments Show p1258 and ps101 cs101?

Applications of microarrays

Microbial gene expression analysis

The predominate application of DNA microarray has been to measure gene expression levels

Profiling in human disease

Arrays have been used to investigate transcriptional profiles associated with human disease and to identify novel disease markers and potential new drug targets **Genotyping**

Microarrays have been widely used as single nucleotide polymorphism (SNP) genotyping platforms

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11. PCR, types and applications?

PCR: Polymerase chain reaction, or PCR, is a laboratory technique used to make multiple copies of a segment of DNA. PCR is very precise and can be used to amplify, or copy, a specific DNA target from a mixture of DNA molecules.
Types: Nested PCR, RT-PCR, qPCR, Multiplex PCR.
Applications of PCR □ PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

12. How polymerase work?

The DNA polymerases are enzymes that create DNA molecules by assembling nucleotides, the building blocks of DNA. ... When creating DNA, DNA polymerase can add free nucleotides only to the 3' end of the newly forming strand. This results in elongation of the newly forming strand in a 5'-3' direction.

siRNA technology
Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing short- term silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation□

Applications of siRNA
siRNA) is the method of choice to target specific genes for silencing and has provided immense potential as therapeutic tools

Applications of PCR
PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.□

PCR-Gene cloning and expression
PCR has been used in gene cloning and screening of genomic libraries□

PCR-Medicine: PCR has major impact on medicine especially in the field of clinical microbiology/diagnosis

Molecular tools have also allowed to perform prenatal genetic diagnosis.□

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PCR-Forensic sciences: Forensic science is the application of scientific procedures to solve criminal & legal matters
Molecular methods are used to establish the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues.

PCR-DNA profiling: DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

PCR-Agricultural sciences and environment
PCR has also facilitated research in detection of pathogens in plants, animals and environment.

PCR-Molecular paleontology
Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains.

Basic PCR reaction:
PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude, generating millions of copies of a particular DNA sequence.

PCR can also be used to generate DNA fragments for gene cloning by using primers with specific restriction site PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude,

generating millions of copies of a particular DNA sequence

▪ PCR amplifies a specific target region of DNA that may range from a few to several kilo base pairs
▪Basic reaction requires several components
▪ PCR cycles may consist of several steps of denaturation, annealing and extension.
▪ DNA polymerases are enzymes that synthesize DNA molecules by adds nucleotide, in building blocks of DNA

Taq DNA polymerase:
▪ Taq polymerase is a thermostable DNA polymerase

DNA polymerases:
isolated from thermophilic bacterium *Thermus aquaticus* and is used in PCR to amply DNA fragments

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Phasmid vectors:

▪ Phasmid vectors are combination of plasmid and ʎ phage sequences

▪ It consists of a plasmid vector carrying a ʎ attachment (ʎatt) site

Bacterial artificial chromosomes (BACs):

Bacterial artificial chromosomes are simply plasmids designed for the cloning of very long segments of DNA (100,000 to 300,000 bp)

Use of pSC101 for cloning

Expression of *Staphylococcus* plasmid genes in *E.*

Yeast artificial chromosomes (YACs):

These vectors replicate in yeast like normal chromosome, but they have sites where very large DNA fragments can be inserted.

YACs with inserts more than 150,000 bp are nearly as stable as normal cellular chromosomes

Plasmid cloning scheme of Ish-Horowicz and Burke (1981)

Cosmid cloning scheme of Bates and Swift (1983)

Comparative genomics of bacteria

Genome of bacteria may be of variable in size. For example, it may vary from 0.49 Mb (*Nanoarchaeum equitans*) to 9.1 Mb (*Bradyrhizobium japonicum* or *Streptomyces*)

Comparative genomics of organelles

Mitochondrial genomes exhibit an amazing structural diversity

Gene transfer has occurred between mtDNA and nuclear DNA

Comparative genomics of eukaryotes

The minimal eukaryotic genome is smaller than many bacterial genomes

Comparative genomics can be used to identify genes and regulatory elements

It gives insight into the evolution of key proteins

OVERVIEW OF CLONING STRATEGIES:

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iii). A means for introducing that recombinant vector into

a host cell wherein it is replicated

i iv). A method of selecting recipient cells that have acquired the recombinant

Introduction:

There are two major strategies for isolating sequences from complex sources

i). Cell based cloning strategy, is to divide the source DNA into manageable fragments and clone everything

▪ Cloning of large DNA fragments of 20 kb in ʎ replacement vectors, fewer clones are required for nearly complete library.

How can appropriately sized random fragments be produced?

i). Average fragment size can be controlled by mechanical shearing

ii). The more commonly used procedure is restriction endonucleases

PCR as an alternative to genomic DNA cloning

PCR with specific primers could be used to isolate genes directly from genomic DNA, obviating the need to produce genomic libraries

cDNA libraries

cDNA library is a combination of cloned cDNA fragments inserted into a collection of host cells

cDNA library is representative of the RNA population from which it was derived

4. type of Resistance observed in p1258 plasmid of staphylococcus (2)?

Use of pSC101 for cloning

Expression of *Staphylococcus* plasmid genes in *E. coli*

pSC101 contain Tetracycline resistance

P1248 from Staphylococcus

13. How animal restore their lost function?

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Screening by functional complementation

Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype.

Functional complementation is also possible in transgenic animals and plants.
It has been used for complementation in transgenic mice to isolate the Shaker-2 gene.

Yeast and Prokaryotic cDNA library?

cDNA libraries are used to express eukaryotic genes in prokaryotes since it does not include introns, and therefore, can be expressed in prokaryotic cells. cDNA libraries remove the large numbers of non-coding regions from the library, and it is also useful for subsequently isolating the gene that codes for that mRNA.

Plasmid stability and copy number?

Plasmid copy number and plasmid stability. Many expression systems in research and industry use plasmids as vectors for the production of recombinant proteins or non-proteinous recombinant substances. ... Related factors are plasmid copy number, structural plasmid stability and segregational plasmid stability.

siRNA Technology?

siRNA technology. RNA interference (RNAi) represents a mechanism invented by nature to protect the genome. ... The underlying molecular mechanism of gene silencing provides us with short interfering RNAs (siRNAs) which allows to target any gene with high specificity and efficiency